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Current legislation and recommendations on ethical post-experimental care and disposition of laboratory animals: a call for global standards

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The fate of laboratory animals once experiments end remains an ethical blind spot. This Comment compares the legislation governing the post-experimental care and disposition of laboratory animals—rehome, reuse, setting free and tissue sharing—across the European Union, USA, UK and Republic of Korea. Highlighting the gaps that persist, particularly in the Republic of Korea, it proposes a decision-making framework and calls on the global scientific community to adopt shared ethical standards for the animals that have contributed to research.

In 1959, William Russell and Rex Burch proposed the groundbreaking 3Rs principle (Replacement, Reduction, Refinement) as a framework to address the pressing ethical and scientific issues surrounding experimental animal use in research¹. Since the introduction of the 3Rs principle, ethical treatment of laboratory animals has become a paramount concern within the global scientific community, transcending disciplinary and geographic borders². Ethical standards for modern experimental animals have primarily dealt with the period leading up to the conclusion of the experiments. However, this narrow perspective overlooks animals that can continue living after the completion of experiments, resulting in an ethical blind spot. Russell and Burch defined the unpleasant mental states experienced by animals as ‘inhumanity’ and explained various experiences, such as fear, conflict and hunger, under the concept of ‘distress’¹. Considering their definition of ‘inhumanity’, the anxiety that animals will face after the experiment ends should also be considered under the concept of ‘responsibility’, aligning with the 3Rs principle that aims to reduce the painful experiences of experimental animals.

Establishing ethical treatment standards for animals that survive the conclusion of experimentation or are not used in studies is a new emerging issue in experimental animal ethics. It is important to expand our social and ethical responsibilities beyond merely determining euthanasia techniques or assuming death as an inevitable outcome¹. We must consider how to responsibly manage the lives of these animals after the initial experimentation. For animals that survive, rehoming should be prioritized as a key ethical alternative to euthanasia. For animals that have already been euthanized, the potential for tissue deposition and sharing should be considered to maximize their contribution to advance scientific

and ethical objectives. Finally, as part of the reduction principle aimed at minimizing the total number of laboratory animals used, the reuse of laboratory animals should also be carefully considered. Despite the existence of legal frameworks addressing post-experimental animal treatment, such as 9 CFR in the US Government³, the Animals (Scientific Procedures) Act 1986 in the UK⁴ and the DIRECTIVE 2010/63/EU in the European Union (EU)⁵, the widespread adoption and standardization of humane post-experimental practices, including rehoming, have yet to be widely adopted within the scientific community. The need for concerted efforts to establish global standards for ethical rehoming and reuse is paramount and is driven by our moral obligation as researchers to ensure the welfare of animals that have contributed to scientific advancement.

Here, we examined the global legal landscape of post-experimental animal care and disposition practices with a focus on reuse, rehoming and tissue-sharing initiatives, not focusing on particular taxa but rather on the sentient animals capable of experiencing pain and distress. We aimed to identify key considerations and challenges in implementing standardized programs by comparing legal frameworks and practices across the EU, USA, UK and Republic of Korea (Korea). Through this comparison, we confirmed that, unlike the EU, USA and UK, Korea still has several legal gaps in its post-experimental animal care and disposition framework. Accordingly, Korea was positioned as a representative example of countries where relevant legal systems remain underdeveloped, serving as a comparative case alongside the EU, USA and UK. Ultimately, this Comment calls for the scientific community to unite in establishing and adhering to standardized ethical practices for laboratory animal reuse, rehoming, tissue sharing and deposition. The aim is not only to enhance animal welfare and promote the reduction principle, but also to reinforce the integrity and ethical standards of scientific research worldwide.

Regulatory frameworks for the rehoming and reuse of laboratory animals

A comparative analysis of the legal regulations concerning the rehoming and reuse of laboratory animals in Korea, the USA, the UK and the EU revealed that all examined countries have established legal provisions regarding rehoming (Supplementary Table 1). However, the regulatory framework in Korea does not specify regulations for the reuse of laboratory animals (Supplementary Table 2). The Animal Protection Act in Korea stipulates that animals may be donated or sold after experiments, provided that their normal recovery is confirmed⁶. Research facilities in the USA are required to maintain records while transporting, selling or disposing of dogs or cats^{3,7}. In the USA, the Animal Welfare Act covers warm-blooded animals such as dogs, cats, nonhuman primates, guinea pigs, hamsters and rabbits used for research, education, testing or exhibition, while excluding rats, mice and birds. Because recordkeeping requirements apply only to

dogs and cats, rehoming practices are effectively limited to these species. Although the target species vary, including dogs, cats, nonhuman primates and ferrets, 16 states have implemented policies that prioritize the rehoming of laboratory animals whenever possible. Various US institutions and organizations such as the National Institutes of Health, American Veterinary Medical Association and American College of Laboratory Animal Medicine^{8–10} have expanded the Code of Federal Regulations (9 CFR)³ to strengthen rehoming policies for research animal disposition. The Animals (Scientific Procedures) Act 1986 (ASPA) in the UK provides detailed regulations and guidelines for the release or rehoming of laboratory animals^{4,11,12}. It mandates consideration of the health status of animals, potential risks to public health and the environment, socialization plans and measures to ensure animal welfare. The EU also permits the release or rehoming of laboratory animals, with considerations similar to the UK.

In addition, both the UK and the EU explicitly recognize the setting free of animals—their return to a suitable habitat or husbandry system appropriate to the species—as a further option for the post-experimental disposition of laboratory animals, a possibility that is particularly relevant to wild-caught individuals (Supplementary Table 1). Under Article 19 of DIRECTIVE 2010/63/EU⁵, setting free is permitted only when the state of health of the animal allows it, when there is no danger to public health, animal health or the environment, and when appropriate measures have been taken to safeguard the animal's well-being. Similarly, the ASPA in the UK⁴ requires that, before a wild-caught animal is set free, consideration be given to its health status, the ecological and predation pressures it may encounter after release, the risk of disease transmission and the impact on the existing ecosystem, as well as to its socialization and the completion of an appropriate rehabilitation program. Although setting free applies to a narrower range of species and studies than rehoming or reuse, its explicit inclusion in legislation underscores that responsible post-experimental disposition should encompass the full spectrum of options, including a return to the wild where this can be achieved safely and humanely.

Regarding the reuse of laboratory animals after the experiments, the Guide for the Care and Use of Laboratory Animals (National Research Council (USA)), while emphasizing the 3Rs principle, does not report or encourage the reuse of laboratory animals to reduce animal numbers¹³. By contrast, the EU not only sets stringent conditions for reuse but also requires institutions to report the number of reused animals, ensuring that such practices are closely monitored and transparently managed. Furthermore, through its Guidance on severity assessment, the EU provides specific criteria for evaluating the cumulative severity that animals may experience over their lifetime when considering reuse, thereby reinforcing a welfare-centered approach. Similarly, the UK stipulates that the reuse of animals is permitted only when authorized under project licenses, provided that a veterinarian confirms full recovery of the animal's health and that the previous procedures were classified as nonrecovery, mild or moderate. In this context, the USA, UK and EU adopt a cautious approach to the reuse of laboratory animals. They all clearly stated that reuse is only possible under strict conditions that prioritize animal welfare and minimize suffering.

In contrast with these governing bodies, Korea lacks any legal provisions to address this issue, highlighting the absence of guidelines that could provide regulatory or advisory support. In Korea, institutions are required to maintain records and documents related to the operation of the Institutional Animal Care and Use Committee (IACUC) for at least 3 years and report the status of animal experiments to the Ministry of Agriculture, Food, and Rural Affairs, including the number and source of laboratory animals used annually. However, while institutions are required to submit annual reports on the number of animals used—classified by procedure severity at the time of the IACUC's proposal evaluation—and their source, there is

no obligation to log in the veterinary record or report the fate of animals after the initial experiments, other than confirming their health status.

Key considerations for the rehoming and reuse of laboratory animals

The regulatory frameworks mentioned above address diverse key elements that should be considered for the rehoming and reuse of laboratory animals. By conceptualizing and summarizing these key elements, we compiled how different countries around the world handle the factors that must be considered in each decision (Supplementary Table 3).

Various institutions consider different criteria as important, such as suitability, suitable adopters, socialization, partnerships and public health (Supplementary Table 3). However, no single criterion is universally emphasized by all institutions, indicating differences among organizations. Specifically, institutions in the USA, such as the National Institutes of Health, American Veterinary Medical Association and American College of Laboratory Animal Medicine, emphasize compliance with relevant laws (federal, state and local laws) and the authority of the Attending Veterinarian (AV)^{8–10}. The UK, EU and the Federation of European Laboratory Animal Science Associations strengthen the role of the Animal Welfare Body (or AWERB in UK) and consider welfare even after adoption, noting that both social environments (contact with humans and animals) and nonsocial environments (visual, auditory and olfactory stimuli) should be considered¹⁴. Notably, the Federation of European Laboratory Animal Science Associations recommendations on rehoming represent the first and only international standard in this area, going beyond existing regulations, and emphasize that adoption programs should account for the total cost of animal welfare before and after rehoming¹⁴.

Based on the summary of key elements that should be considered for reuse (Supplementary Table 3), it is evident that organizations and governments mention fewer considerations for reuse compared to rehoming. According to the Animals (Scientific Procedures) Act 1986 of the UK⁴ and the DIRECTIVE 2010/63/EU of the EU⁵, animal reuse in new procedures is permitted only if four conditions are met: (1) severity of previous procedures, (2) veterinary confirmation, (3) health condition of the animal and (4) severity of further procedures. These regulations permit reuse only under stringent criteria to effectively manage and regulate the reuse of animals while limiting their prolonged suffering and ensuring adequate welfare.

Regulatory frameworks of deposition and sharing of laboratory animals' organs and tissues

A comparison was made between the regulatory frameworks related to the deposition and sharing of organs and tissues of laboratory animals after the experiments in Korea, the UK and the EU (Supplementary Table 4). While the EU and UK provide detailed regulations on deposition, the US lacks clear regulations on tissue deposition, including those in the Guide for the Care and Use of Laboratory Animals, and cannot be compared with the other countries in this regard (Supplementary Table 4).

Article 5 of the Laboratory Animals Act of Korea¹⁵ stipulates the operation of a Laboratory Animal Resource Bank for the preservation and management of resources derived from laboratory animals. The 'Enforcement Decree of the National Research and Development Innovation Act' and the 'Act on the Acquisition, Management, and Utilization of Biological Research Resources' mandate the deposit and preservation of biological research resources obtained through government-supported research, aiming for systematic resource management¹⁶. However, the implementation of these systems is not mandated for inclusion in the IACUC review or in animal experimental planning, resulting in a lack of practical regulations or operational standards.

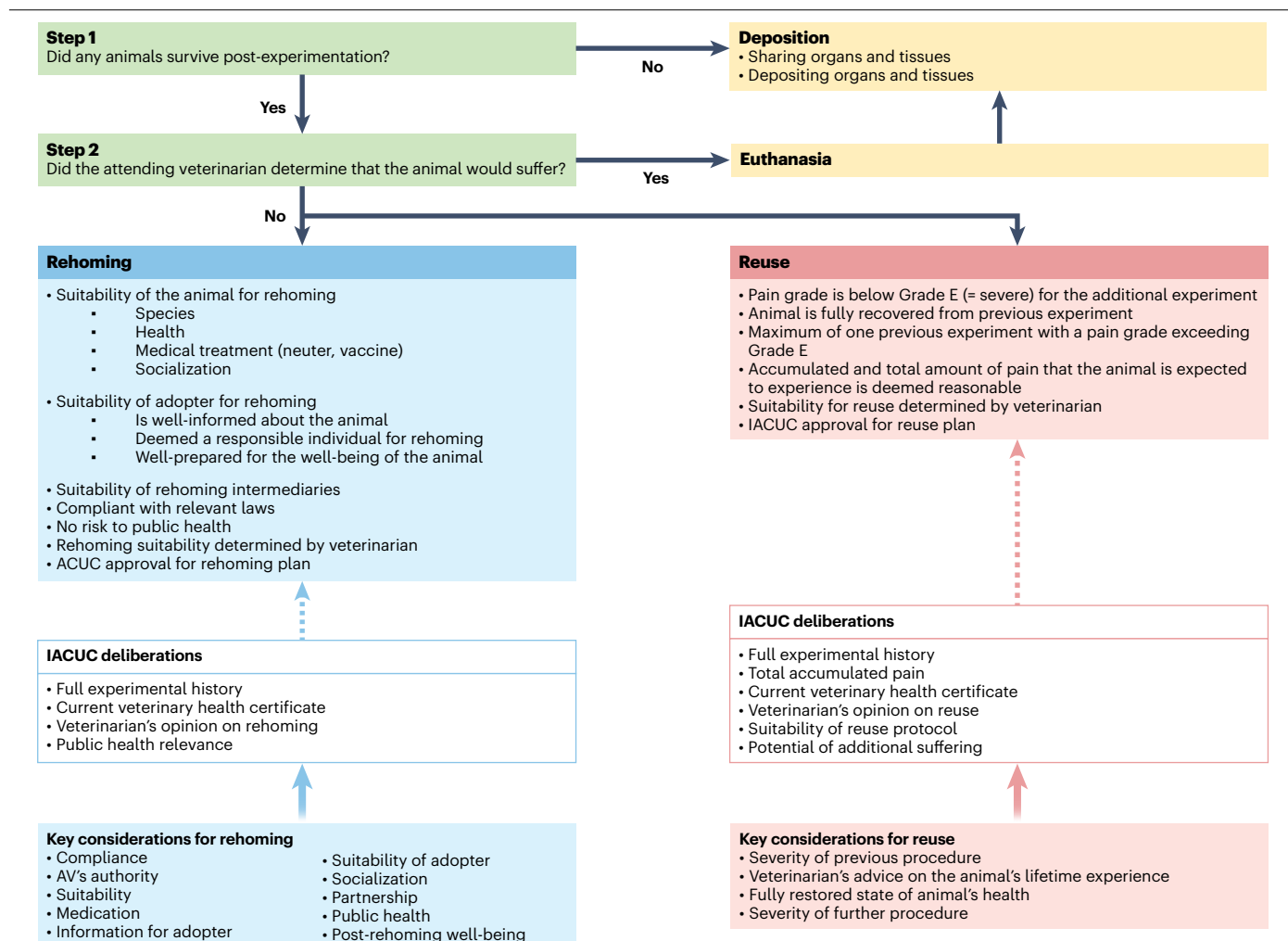


Fig. 1 | Decision-tree proposal for the post-experimental animal care and disposal plan, with the IACUC as the final decision-making body. The flowchart guides researchers and the IACUC through the options available for animals remaining after a study. Surviving animals may be considered for rehoming, reuse or setting free, whereas the tissues and organs of euthanized

animals may be deposited or shared. For each pathway, the final decision rests with the IACUC, based on the Attending Veterinarian (AV)'s assessment that the animal will not experience pain or distress, and on consideration of the animal's health status, the severity of previous procedures and the relevant legal requirements.

The UK goes further by stipulating the establishment of programs for sharing organs and tissues from deceased animals for scientific purposes as a responsibility of the Secretary of State⁴. In addition, as part of the government policy to reduce animal use, the sharing of research results, resources, animals and data is actively encouraged. The EU also requires each nation to facilitate the establishment of programs to share organs and tissues from deceased animals⁵. To this end, the UK operates various resource-sharing programs led by the NC3Rs, and the laws of the EU and UK explicitly require governments to support the establishment of organ- and tissue-sharing programs for euthanized animals¹¹.

Global standards for ethical laboratory animal rehoming and reuse: a call for action

Regarding rehoming, reuse and deposition of laboratory animals, the legislative framework of Korea was found to be lacking compared to the UK, USA and EU, which are considered to have advanced systems. To tackle this situation in Korea and improve experimental animal ethics, institutional

improvements are necessary. Even though the Korean government has recently made various efforts through legal amendments to address these gaps, beyond regulatory improvement, fostering a culture of ethical responsibility within the research community is equally essential. Researchers and institutions should take the lead in developing voluntary standards and best practices that go beyond legal requirements, reflecting a shared commitment to the welfare of laboratory animals and the integrity of science.

Regarding rehoming, there is a need to raise awareness about the life and suffering experienced by surviving animals post-experiment and establish clear systems to help researchers and institutions realistically implement adoption, including for rodents and wild species like nonhuman primates. Guidelines should also be established to ensure animal welfare during the gap period between protocol approvals. In particular, when considering reuse as an option for surviving individuals, there should be strict evaluation criteria that comprehensively consider the possibility of animal survival, previous research history, current health status and veterinary opinion. To efficiently manage deposition of euthanized individuals,

concrete regulations are required for the storage, sharing and disposal of animal tissues after experiments. In addition, an international network should be established to promote ethical considerations, data management and collaboration among researchers in the tissue deposition process.

The fate of animals after the experimentation should follow ethical and rational standards, necessitating the establishment of a globally applicable framework and subsequent discussions. As discussed in detail in the case of Korea, various countries have substantial differences in the specificity or existence of regulations related to animal welfare. Therefore, establishing consistent standards for animal treatment and creating international discussion forums are important starting points. When researchers use a decision-making framework to help them plan specifically and carefully for the fate of animals after animal experiments, they could consider animal numbers and post-experimental handling plans in their design from the experimental planning stage and make ethical choices.

This Comment aims to present a decision-making framework that includes all the key components of laws and guidelines from selected countries (Fig. 1 and Supplementary Fig. 1). As specifically explained in Fig. 1, living animals that remain after the experiment can be rehomed or reused. The final decision on whether the remaining animals may be rehomed or reused is made by the IACUC, based on the veterinarian's assessment that the animals will not experience pain or distress. Throughout this process, researchers should be able to provide all relevant information and explanations, as requested by the IACUC, which is critical for ensuring transparency and accountability. Careful calculation of the number of offspring of the experimental animals (Supplementary Fig. 1) should be performed before following the proposed decision tree (Fig. 1). Certain countries commonly emphasize that researchers have an ethical obligation to establish responsible plans for the production and handling of animals to be used in experiments^{4,5,8}. This includes breeding only the minimum number of animals necessary and achieving research objectives through optimal management of bred animals. During the IACUC approval process, the management plan for experimental animals awaiting approval for rehoming or reuse should be clearly defined, and humane endpoints should be set to ensure that animals do not experience additional pain during this management period (Supplementary Table 5).

In conclusion, the comparison of laboratory animal-related regulations from various countries revealed differences in animal welfare approaches and research ethics across those countries. To fulfill ethical responsibilities in animal experimentation, the systematic establishment of relevant laws and guidelines is essential in each country. The enactment of animal welfare laws, advancement of researchers' awareness and judgment, and systematization of the IACUC are not issues specific to any one country but need to be improved globally. When institutions in each country consider using forms such as those shown in Fig. 1, researchers can more easily establish ethical decision-making processes. Ultimately, by developing a global framework to ensure ethical animal experimentation and promote ethical awareness, a sustainable research environment that respects animal rights and welfare could be achieved.

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Additional information

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